

ActInf MathStream 007.1 ~ "Constructor Theory as Process Theory", Vincent Wang

MathStream | Vincent Wang-Mascianica et al. | 1:36:38

Hello and welcome everyone. It is September 19th, 2023, and we are in ACTIMF Math Stream number 7.1 at the Active Inference Institute. We're here with Vincent Wang Mashinitsa and guests.

It should be quite a discussion, certainly a unique one by what we can already see. So thank you all for joining. We'll be looking forward to this presentation and interleaving discussion.

Thanks, Dan. So hi, my name is Vincent and with me today also is Hamza in the audience.

We're both authors on this paper that we'll be talking today, Constructor Theory as Process Theory.

So what should the format of this discussion be?

So I'd like to tell you a bit about Constructor Theory and Process Theory and then sort of have a discussion about the sort of diagrammatics of Process Theory, how that might fit with Active Inference.

And all this, you know, you can view it as an exercise and sort of seeing things in a diagrammatic way.

So I'll get started and the structure will be roughly I'll talk about what Constructor Theory is first, just from a conceptual level, because I'm the only non physicist among the authors of this paper.

Hamza is a physicist, so any questions about specific examples is there to be directed to Hamza.

And then, you know, after that, I'm going to do just a sort of lightning fast introduction to what a process theory is.

And then we'll put the two together and then we'll have a discussion.

And throughout, I will be making pauses to sort of invite questions and make sure everyone is on the same page conceptually.

But everything should be super, super easy, because I think all the ideas really are, you know, fundamentally, they're quite simple.

Okay, so without further ado, what is Constructor Theory?

Constructor.

So you may have heard of this already, because there's quite a lot of hype.

They have a very good PR machine, if nothing else.

So among other things, it's been described as, and I think this is from a Quanta article, a master theory, a set of ideas so fundamental that all other theories would spring from it.

In fact, if you go to Deutsch's original paper on Constructor Theory, he goes so far to say that all other theories of physics, you know, we'll call them, you know, it doesn't matter if it's thermodynamics or quantum or relativity, gravity, whatever.

We'll just call them subsidiary theories, because that's how big and important Constructor Theory is.

And, you know, there's this eponymous book by Chiara Moletto, the science of cannon cats.

I want to say that, oh, well, you know, Constructor Theory is about counterfactuals and what is and isn't possible in the domain of physics.

And all this is all this sounds really impressive.

But, you know, the question is, what do you mean by what do you mean by this?

So how this how this all started is some of Chiara's students, Anisette and Rhea Villarres, they came over to

the Wolfson Quantum Foundations discussion.

And they were nice enough to sit down and tell us about what Constructor Theory really is, because, you know, historically, it's been really difficult to sort of pin down, you know, to pin down like, what is the math?

What do you really mean here by all these words?

And, well, you know, they're happy to say, look, you know, it's a meta theory, you can sort of bring your own mathematics, you know, what we really there's a sort of methodological approach of Constructor Theory.

That's more important than the specific mathematics that you want to use to instantiate all these words, like what is a Constructor, what is possible, what is impossible.

But, okay, I'll tell you conceptually what Constructor Theory is.

So first, I want you to, let's talk about a space of conceivable processes.

Conceivable is the word here.

Conceivable is the word conceivable.

Conceivable will become clear soon, because we're going to cut this space shortly.

And what are the sorts of conceivable processes that we might have?

Well, let's look at a limited subset.

Let's talk about ice cream.

There's sort of, there's a sort of simple process that can always happen, which is you might have some ice cream, you leave it out for too long, and then the ice cream melts, becomes a puddle.

This is, you know, we want to say, this is obviously a conceivable process.

There's another conceivable process where the opposite thing happens.

You have the ice cream that's melted, and then it spontaneously unmelts and becomes an ice cream again.

This is a conceivable process as well.

And then there's the straight up, you know, ridiculous stuff that, you know, we know for sure is not going to happen in the domain of physics.

Like, for example, you have an ice cream, and then you've got some machine such that you can take it and you can take, you put in anything and it performs, it performs this process or task of taking your input and copying it, everything perfectly down to the, down to the quantum states.

You know, you're cloning the ice cream, maybe where there's this quantum no cloning.

So we know that this sort of thing is, you know, not going to happen.

Nevertheless, all of these are conceivable processes of relationships between possible inputs and possible outputs.

I should stop using the word possible, just inputs and outputs.

Right.

So what is the aim of constructor theory really?

Well, it stems from this desire to classify the space of conceivable processes into that which is possible and that which is impossible.

Well, what might we say intuitively?

We might say that, well, we're pretty sure that we're pretty sure that this ice cream melting stuff, that's possible in terms of some theory of physics that explains everything.

And I go, okay, fine.

Like, some theory of physics that explains what happens to ice cream definitely want to say that ice cream melting is possible.

And we definitely, you know, that theory of physics also has got to say that this sort of cloning situation here, that's got to be, that's got to be no good.

That's, that's impossible.

And then the difficulty comes when you've got the cases in the middle where, okay, well, technically, if we're just using English, I mean, it is conceivable, it is possible that an ice cream can spontaneously reform itself.

I mean, it can happen.

The laws of physics don't forbid it.

It is simply very, very unlikely.

But it's Deutsche's idea that we should say, no, no, no, you know, let's do away with the complexities of statistics and just say that that's straight up impossible.

Right.

So then the goal of constructive theory is to take the space of conceivable processes and cut it.

Just one cut corresponding to on one side, what is impossible.

And on the other, what is possible.

Right.

So just using this one sort of concept or duality, what is possible and impossible.

It is like you take your space of conceivable processes and you just determine what's possible and impossible.

And here's the subtle part.

And here's the subtle part.

Then you claim that any theory of physics corresponds precisely to the content of one of these cuts.

And what do I mean by that?

Well, I mean, is that, well, you know, here's this is one space of conceivable processes that happens to talk about, for example, the dynamics of ice cream.

But there are all kinds of conceivable spaces of processes that might govern, you know, different sort of domain or domains or magisteria or whatever you want to call them.

Of phenomena.

And the idea is that, well, you know, anytime you have a theory of physics, what do you really have?

You really have just the domain of phenomena that you care about.

And constructive theory says, and in addition to that, a cut that tells you whether, you know, what things are possible in that domain and what things are not possible.

And so then every pair of, you know, sort of domain plus one of these cuts that tells you what is possible and impossible is going to kind of encode or summarize just be a particular theory of physics.

Right.

And so we can be very empathetic to this.

We can be very empathetic to this kind of formulation.

It's a nice thing to try to do to sort of to summarize or have a sort of bird's eye view of all these different theories of physics to try to boil it down to something that's conceptually a bit simple.

But of course, there are problems, too.

Right.

So one problem is, well, you know, why didn't you call it a possibility theory rather than constructive theory?

I don't know the answer to that.

But possibility theory sounds a bit too much like a mathematical theory, and it doesn't sound as sexy as constructive theory.

So that's one reason you might want to call it constructive theory.

But then what is a constructor?

What do constructors have to do with all of this, this idea of possibility and impossibility?

Well, you see, the difficulty was if you go back and you think about the case of the ice cream melting.

That's something that is possible.

By our usage of the word.

But you'd want to be able to stick it on one side of the cut and call that impossible.

Right.

The solution, unfortunately, is verbicide.

Verbicide, what you do is you simply make up a technical concept.

You call this concept the possible you call this concept possibility of a task, let's say.

And then you.

Well, what do you mean by possibility of task?

Well, that relies on, you know, defining what these constructors are.

Possibility defines depends on defining what constructors are and constructors in turn depend on on the definition of tasks.

And then tasks, finally, are something that you can cash out in terms of processes.

So this is this is the idea of where constructive theory wants to go.

And now I'm going to tell you how it is that constructive theorists get there.

Right.

Let's see.

Do we have any questions at this point?

Any sort of conceptual questions or clarifications that anyone wants to?

Because otherwise I'm going to move on.

Where does likely or unlikely fit?

That's all on one side of the line.

What if something's vanishingly unlikely but possible?

That is an absolutely brilliant question.

And the answer is.

You get to choose as long as you make a cut, because so the dogma of constructive theory goes, you only

get possible and impossible.

I don't care how you pick how to distinguish the likely of vanishing unlikely.

And I didn't make up the rules.

I'm merely telling you that everything has to be either possible or impossible.

Good question.

Sorry.

That's the best I can do.

Okay.

Three.

So.

Right.

What is constructor theory?

All right.

It's a.

It's a meta theory.

Bring your own math.

The meta theory.

What does that mean?

That means bring your own math.

Okay.

Let's start from.

What is a constructor?

Well.

You know what?

This might be.

This might be better.

If I.

You know, I'm looking at.

I'm looking at these notes now.

And I'm going to.

Nope.

I'm going to.

It wouldn't make sense if I started from there.

I'm going to have to put that as the as the bottom thing.

And then we'll get there.

What is.

Constructor.

Right.

Okay.

Let me start with defining what a task is.

So all of this so far is going to be in English.

And then we're going to translate to math in a second.

Hold on.

Let's.

Let's talk about what a task is first.

From a task.

You get on to.

This.

Remember, we said that something was conceivable.

We have to break down that initial diagram that we drew.

This initial informal thing.

We had the sort of the space of all conceivable tasks.

So first we have to define.

What do we mean by tasks?

What do we mean by conceivable tasks?

And then from conceivable, let's just talk about what do we mean by a possible task.

Right.

And then it will turn out that a constructor is a.

The constructor is.

Is something that's associated with a possible task.

So that you see that there's a difficulty here of getting from constructors to.

You know where ultimately you want to go.

There's.

There's a sort of.

Yeah.

Right.

Logic of justification, logic of discovery.

And they don't know.

They don't know.

They don't know.

They don't know.

Okay.

Fine.

So what is a task?

Well, remember, we're allowed to.

We're allowed to bring our own mathematics here.

So we asked them.

We took a lot of care to ask them, you know, what sort of math do you want?

You know, what do you need here and there?

And then finally, you know, it came down to look, look, really what we care about as constructor theorists is just defining these things.

Once you pick some sort of sensible interpretation of these things in mathematics, then you can go ahead and play the game of constructor theory.

So we go, okay, that's fantastic.

You know, so here we have a here in Oxford, we have a we have a nice sort of homebrew math.

What we, you know, we call them process theories, but otherwise they're known as symmetric monoidal categories.

We say, okay, well, let's let's call let's call a task a process.

A process.

In a symmetric monoidal category.

Okay, well, we'll get to what that means in a second.

So what is a conceivable task that becomes?

Well, that becomes mathematically trivial.

It just becomes a process inside a process theory.

Okay.

What is a possible task?

Well, possible task is going to be.

And then here's where we have to get technical again.

What is a possible task?

Well, it's defined as an infinitely iterable task.

Infinitely iterable.

Yada, yada.

Okay.

And then if a task turns out to be one of these infinitely, if a task turns out to be infinitely iterable, we call it possible.

And a task always is doing something and then there's something in the environment that's helping it do it.

That's something in the environment.

The super ancillary thing at the end of these definitions is going to be what a constructor is.

And at this point, it will be very helpful to actually go through an example.

And it so happens that by going through this example, just in the diagrammatic form, you will understand process theories because the whole point of process theories and symmetric monoidal categories is that you have this formal diagrammatic syntax that's extremely intuitive.

You don't have to know anything about the category theory.

You can just use the diagrams and it's a formal syntax where you sort of kept all wrong as long as you're just like composing the boxes and the wires.

So I'm going to now give an example of, I'm going to make an example of a constructor.

And this is also a good time.

So now that I'm erasing, anytime I'm erasing, it's a good time to chime in with questions.

Sure.

One question from the chat.

Upcycle Club wrote, does this mean that constructor theory as a process theory indeed takes into account the no deleting theorem?

No deleting?

What do you mean?

I'm not familiar with this.

Not sure if that's what was written.

Or maybe you referenced no cloning earlier.

I referenced no cloning.

Ah, so here's the nice bit.

So earlier I drew a diagram where there are all these different blobs of different colors and each one was a different domain of physics or whatever.

Right.

So it turns out that, you know, part of the content of a process theory or of a symmetric model category is that you have to pick a category, an ambient category in which you're interpreting.

And that choice of ambient category tells you about the sorts.

That is a choice of the conceivable processes.

So if you choose the category of quantum processes and quantum maps, it will turn out that in that setting you have no cloning.

But if you pick a different setting, like for example, maybe you want just sets and functions, you know, mathematical sets and functions, in which case, of course, you can copy.

You can always, you know, copy down any information that you have, you know, as long as it's all on paper and you're just using sets and functions, right?

But in physics, fine, you want to model it with the process theory of quantum maps, then you don't get it.

So it depends on the choice of interpreting category.

But that's a very good conceptual question.

Thank you.

Right.

So talk about a task.

I'm going to make a task here.

I'll draw these tasks as directed boxes with wires.

So this is just, what am I drawing here?

Okay, I've drawn these directed because I want everything to be read from left to right.

Okay.

Now, in the process theory, every process is drawn as a box.

A process is something that takes as input over here, some systems and does something to it, whatever.

Then it outputs some systems.

And then processes can be composed in, you know, kind of usual way that one expects.

This is a composite process, for example.

So what is the process?

So what is the process here?

Well, okay.

Let's talk about these systems.

What do I mean by these systems?

They can be just about anything.

They're subject to a couple of rules.

The rules are, anytime you have a system, you just need some way to talk about what it means to have two of these systems side by side.

So let's say that the green system is, see the green system here.

So now this is something that takes in a green system and an orange system and gives back the green system and the orange system.

So it's something that's acting on these systems.

It returns them.

What do I mean by a system?

Well, a system can be, well, mathematically, we would call them a type.

Like a system can be like the Booleans or they can be a set or they can even have the vector spaces as systems.

And if you have vector spaces as systems, there are lots of different ways that you can interpret this idea of sticking to systems side by side.

You can have the direct sum of vector spaces.

You can have the tensor product of vector spaces.

And all of them would give you valid process theories.

So you have linear maps with direct sum or linear maps with tensor products.

And that would all give you a process theory or asymmetric nodal category in which you can interpret the diagrams you're drawing here.

Let's, let's come up with a simple one though.

A simple one is let's say that this green system is going to be, oh, this is terribly drawn.

That's a slightly better.

It's shoes.

All right.

It's a shoe.

Let's say it's a particular shoe, right?

That, that helps.

And what is this other system?

So what is this other system?

I'm going to call this other system, the system of a bucket with a paintbrush.

Bucket.

And here's a paintbrush.

Right.

This is a perfectly fine systems.

Talk about.

These systems can be in different states.

Like for example, a shoe can be, well, it can be clean or it can be dirty or have all these different colors.

With the bucket, well, let's, let's talk about buckets, you know, sort of having paint inside.

Maybe the bucket has some paint in it.

Maybe it doesn't.

That's a good example of the system.

And I want to model a particular task here, which is paint the shoe.

Right.

It's a process.

Why do I call it a task?

Well, to turn it, to, to view a process as a task, I have to, I have to look at it in the right way.

I have to say like, okay, well, there's gotta be one system.

There's gotta be one sort of system.

Let's input and output.

I want to call that system, the substrate.

Well, why?

Because the task has to know that that's the substrate there.

The, you know, painting has to happen to something.

If you're one of you painting as a task in the substrate of the painting is going to be the shoe.

And what about this other, what about the, what about the paint?

[illegible]

If anyone's unhappy or has any conceptual questions, and now's the right, now's the right time to ask.
It seems like we kind of put a box around the entire crux of the issue, but at this level of abstraction, indeed.
Right.
Indeed.
Right.
I mean, like, yeah, of course.
There's a lot of boxing away the crux of the issue, but in fact, we haven't even gotten to the crux of what a constructor is yet.
I'm merely defining what it is for something to be a task.
I mean, you can then pick.
So the point also of a process theory is that you can work at the level of abstraction you so choose.
Think of this, you know, if you code, this is basically you're declaring what the types of the method or the function are.
It's still on you to go fill in the implementation details.
But here we're just dealing with the level of this.
This is the level of abstraction where it's perfectly okay to start working to just define what we mean by a constructor.
And remember that this is still just an example.
At some point, you can just erase all the labels and just keep the coloring that indicates the types.
And then that would this would be the abstract specification of what a task and a constructor is.
Yeah.
So you're right.
We are hiding away a lot of the we are hiding away, you know, the crux of the difficulty or the meat.
But we're doing so in service of sort of getting at a formal incarnation of what constructor theory is.
So, yeah.
Okay.
So.
Kind of happy with this.
Now I'm going to.
Reduce this.
Let's see.
I want to hear.
So now, well, let's suppose that it's, you know, maybe it's a small bucket of paint.
It's a fairly big shoe.
Well, either case, the content of the equation was that, you know, you stick in this full bucket of paint.
And then at the end of it, you run out of paint.
You finish painting a shoe, but you run out of paint.
And what that means as a sort of diagrammatic consequence.

What happens if we want to paint a shoe twice?

Well, I mean by that.

Okay.

Well, here's two shoes.

And just to distinguish them, I mean, just to distinguish them, let's say that this one starts off like a nice yellow shoe and this one's a nice orange shoe.

And then in the end, we still have this, you have this again, this bucket of paint.

That is kind of, that is full.

What do I mean by painting, painting two shoes?

Well, okay.

Let's, let's, let's paint them in the order that they're closest.

So first we want to apply the paint shoe task.

Right.

We end up with some shoe system.

We end up with some bucket of paint system.

We want to say, okay, we'll send that, that first system.

We send it away.

That second system, we'll bring it in.

Right.

Now let's line things up again.

And let's apply this again.

Right.

Here's a paint shoe.

And here's, you know, let's try it again.

Let's try to paint the shoe.

Right.

And now we've, we're, we're seeing a new diagrammatic component, which corresponds to the word symmetric in a symmetric monoidal category.

The idea is that, well, in a symmetric monoidal category, yes, you can put systems side by side, but in a sense, you don't care about the spatial configuration of how you do it.

You're always free.

You sort of have a bag of systems and you're allowed to pick them out.

And geometrically, what this amounts to in the diagrammatic syntax is saying that, well, look, if you have a system side by side, you can always, you can always swap the two systems and you can depict the swaps as, as wires twisting over each other.

And the sort of the difficulty of, or the only difficult content of a symmetric monoidal category is showing that, you know, the, the way that the diagrams behave topologically correspond precisely to the algebraic content of, of the morphisms that are the semantics of the, of the processes.

Which means, which, which, which is to say that as long as you're tracking the connectivity and the

connectivity matches between two different diagrams, they're going to refer to the same formal entity.
So there, so it's really like a safe syntax to, to use.

Right.

But we can follow this, right.

And this is the, this is, this is what I said.

I'm, I'm painting, I'm trying to paint the two shoes with one bucket of paint.

That should be clear.

I'm trying to arrange to paint the first shoe, the orange shoe.

And then I twist in the second yellow shoe and try to try to apply that process again to do the painting.

Right.

And then remember the diagrammatic equations, the diagrammatic equations let us reason directly with this.

We never have to sort of leave the diagrams once we've defined everything.

So what do we have?

Well, okay.

The diagrammatic equation tells us.

Of what painting shoes does is okay.

Well, this is going to be then the same as.

Oh, I empty the bucket of paint.

I paint this blue.

Right.

And you'll note that, you know, it doesn't matter whether I stretch things.

It doesn't matter if I stretch things or I twist the wires around in a symmetric monoidal category.

It, you know, everything up to stretching doesn't matter.

All refers to the same configuration of states and applied processes.

But then I run into a problem here because, okay.

So in a symmetric monoidal category, everything is okay up to twist.

So this thing over here that crosses over, I can sort of pull it.

I can yank it all the way.

Okay.

To here.

Right.

And then what do I have?

Well, what is this saying?

I've just done these diagrammatic derivations to show that, well, I applied the equation that tells me that I can use a full can of paint to paint a shoe.

And then I run out of paint.

And now I'm trying to apply an empty can of paint to a shoe that needs to be painted.

At this point, I'm stuck.

I have no more diagrammatic equations to help me, you know, get this into a simplified state from here. Because, well, we might want to define the painting shoe process to be one such that nothing happens. If you put in the empty bucket of paint, you don't end up painting the shoe.

Okay.

For this reason,

not constructors yet.

Let us imagine, then, that we had this magic bucket of paint.

Now this is, I've cast a spell on it.

This is a magic bucket of infinite paint.

And the nature of this infinite paint is that it's always going to be full.

It's always going to have enough paint to paint another shoe.

Of course, such a shoe, such a, such a bucket of paint cannot exist in our physical reality.

You always have to use up some paint.

But if you had this sort of mathematically ideal infinite bucket of paint, then it doesn't matter how many shoes you try to paint.

You could always then sort of apply another equation.

And then you go like, well, you know, the infinite bucket of paint is not going to run out.

So I can, I can apply the equation again.

And I can paint another shoe and I can keep painting shoes.

And every time I apply the equation, well, it never goes from a full state to an empty state.

It just stays in the same infinite state.

And it allows me to perform this task again and again and again, however many times I want.

Rather counterintuitively, this is what is meant by possible.

So painting a shoe, as I've described it, using just a regular bucket of paint that empties, is actually not a possible task in the framework of constructive theory.

Right?

This is what I mean by verbicide.

You know, you'd want to say, of course you can paint a shoe.

No, no, no, no, no.

Possibility is a completely technical term that refers to the ability of a task to be iterated infinitely.

And then this thing, this infinite bucket of paint here, this state that enables the task, the environmental state that enables the task to be repeated infinitely, which is to say, after you apply the task once, the resulting transformation on the environment leaves it in such a state that it can do the task again.

That thing is a constructor.

Okay.

Okay.

So, we're going to move on to what's next.

What is next?

What is next?

But this is a good time to ask questions.

This is a very good time to ask questions.

Mohammed, do you want to ask a question?

No, I don't have any questions.

Okay.

Well, why?

I suppose, what else can I say about constructors here?

Well, conceptually, where do constructors come from?

If you think about catalysts for some biological reactions, the notion of a catalyst in chemistry and biochemistry, this is one of the sort of intellectual sources of this idea of a constructor.

If you replace constructor with catalyst, it doesn't really change much about how you should intuit the thing.

It's this idea that, well, you need it in order to help this transformation occur, but then at the end, you just get the catalyst back and then the catalyst is unchanged and you can go and participate in another reaction.

So, this is the sort of idea of what a constructor is.

Okay.

So, why again is, what should I talk about here?

We can talk about, okay, we still have to get to this idea of the cut.

So, how is it that having this notion now of a constructor helps us define what we mean by a cut?

Yeah, that's it.

Cut of possible and not possible.

Okay.

So, what is a constructor?

Okay.

Okay.

Let's, I'm going to rehearse one more time this, but I'm going to simplify it.

What do I want to do here?

Yeah, okay.

And I have now this, this infinite bucket of paint, right?

And I have, well, the shoes I'll get to in a second.

Okay.

Okay.

So, for whatever reason, constructive theorists really like to use the category of sets and relations.

They really like to think of things in terms of sets and relations, which is a problem when you're trying to model quantum things.

We can get into that in a second.

How did this come about?

Well, you know, the usual way, which is to say when David Deutsch wrote his first paper, he said, well, you

know, let's just, you know, let's take sets and relations as this sort of setting where we can start doing some calculations.

And we'll use it as a placeholder setting for now.

And of course, you know, no one, no one bothers to ever go back and change the placeholder.

Then the placeholder becomes a standard.

So, okay, fine.

They use, they use sets and relations.

And then, so the, the systems in the process theory, we'd call them systems, but in the constructor theory, we'd call them substrates.

So the environments, they're all just going to be sets.

They're going to be sets of, they're going to be sets of, and again, we're, we're, we're dying under the terminology here.

Sets of macro states.

Why do we want, let's, let's just talk about macro states.

Well, okay.

I mean, like this is a, this is a deep constructor theory lore thing because they need to just, they need to talk about like thermodynamics in a second.

So why macro states?

Because there's lots of different ways a shoe can be blue.

You know, you know, if you're really looking at it at the, the molecular level, you know, there's all these different configurations and, you know, you kind of don't care.

And you want to say like, well, look, all of them just count as blue.

As far as I'm concerned, we'll call all of that.

We'll call all of that.

The, the macro states of the, the shoe being blue.

Then there's shoe being whatever other color, and you don't really care.

And then the job of this task, job of this task here.

Is just to take.

The macro states that are not blue and to turn them into macro states that are blue.

And it must be so for.

Well, it doesn't matter if this is a set that's going to be a set of.

Micro states, macro states.

Turns out doesn't matter when you're working process theoretically.

We show in the paper that like, look, there is this, there is this distinction at the mathematical level when constructive theorists talk about this between like a particular molecular configuration and the macro states of the collection, a sub collection of configurations that are blue or whatever.

But, you know, what we show is that look, look, in this system, it doesn't matter.

You can just work directly with the macro states because that would, that's what you really want.

And it turns out mathematically doesn't make a difference.

So what are we saying a constructor is?

So here's the definition of a constructor.

We're saying that.

This equation ought to hold.

Right.

What do I mean by this?

I mean by this.

So in the language of, in the diagrammatic language for sets and relations, you've got a special state and you've got a special something that's the co-state or in effect, something that only takes inputs and doesn't give you any outputs.

What's this special state?

Remember states in the symmetric model category of relations, set and relations, are subsets.

And this special state is the special subset of everything in the, so what this part of the diagram says here, just up to here, saying, okay, well, look, here's my constructor.

That's in my environment.

And my substrate is, well, okay, I don't care what you put in here.

It could be any macro state.

But you perform this task.

And then the opposite of giving me everything.

Well, here is the effect, the relation that is the converse of this is, well, something that takes every, it's something that takes every element of the set to the single singleton.

So it's, it's the equivalent of deleting or forgetting about it.

And in total, what does this say?

Here, look, if you apply this, if you apply this process, view it as a task, it takes this constructor in the environment acting on a substrate.

Now, look, if you, if you forget about it, marginalize over, you just trace out or whatever words you want to use to say, just forget about it.

It doesn't matter what you put in, in the substrate.

You're going to get back something that is good to go again as a constructor.

And now, rather than equality, because of working in sets and relations, and recall that these states are now subsets.

We can say, well, look, if I, if I give you, if I give you a constructor, what I, what I mean to give you here is a subs, I'm going to give you a, a set or a collection of macro states that all work as constructors.

Because of course, the infinite bucket of paint might also have micro states, you know, the paint could be in different configurations, but they might all be equally infinite and good for the product, good for this task.

But I give you all of these, and if you apply the, if you apply the process to it and you do the task, you get back, you get back a set of states that is good to go again as, to run this, you can just plug that into another copy of this equation and you can just get it again and again.

So, this is now minus the, now we can just take away all of the sort of stuff that's still tethering us to intuition.

And at that point, that is just the definition of a constructor.

Right, this is, that's a constructor.

Okay.

So, what can we do?

What can we do with,
constructors?

So, recall now, this is, this is going to be the constructor, and this is by virtue, a task, which by virtue of having a constructor, that's going to be a possible task.

Right.

Possible task.

And now,

the final bit that we need to get to, to end the, this conceptual story of what's going on in the paper,
and we can move on to a discussion,
is,

well, what did, what do we mean by this cut?

Remember, there was this cut between, you know, so there's all these conceivable tasks,
and then there was a cut,
then you had all these possible tasks,
and then all these impossible tasks.

How do you get that cut again?

Well,

you know, you have a sub-process theory.

So, process theory,

process theory is closed under composition in two ways.

You can compose processes in sequence,
and you can compose processes in parallel.

A process theory says, well, you give me a bunch of processes,
and a process theory says, well, I'll give you the closure,
all conceivability is the closure of these processes under sequential and parallel composition.

And just, you know,

what do I mean by sub-process theory?

Well,

in the same way that you can have a subgroup,
or you can have a,
you can have a vector space that embeds into another,
you know,
a sub something of some other mathematical construct,
we're saying,

well,

you can also have another process theory where you start off with a subset of processes,
and then you take the closure of that under sequential and parallel composition,
and twists in the case of the symmetric modal category.

And then you end up with,

well,

that's also a process theory,

and it can be a process theory that's a subset of a larger process theory.

And so,

translating all of that into,

so,

translating the sort of mathematics there into sort of touching the constructive theory,

we'd be saying that,

look,

what we would have to show in order to claim that,

you know,

this definition is somehow good for where constructive theory,

constructive theorists want to go,

is we want to say that the possible tasks themselves form a process theory.

The possible tasks,

you know,

what do I mean by that?

Well,

that means just that,

you know,

if I give you or identify where the possible tasks are,

which tasks are possible in an ambient process theory,

then these possible tasks with constructors form a sub-symmetric monoidal category,

which is to say,

the sequential or parallel composition of possible tasks,

again,

yields a possible task.

Right.

And that is not too difficult to show.

That is not too difficult to show.

And perhaps like you can,

you know,

this is the sort of thing that you can,

this is the sort of thing that you can try at home.

If you have,

if you have the pen and paper,

follow along.

All I have to do,

well,

once you have this equation,

I mean,

all you have to do is check,

you know,

because this is,

this,

sorry,

this in equation,

this subset hood relation,

must be verified to define whether a task is possible.

This,

this inequality simultaneously defines possible tasks and constructors.

So that has to be verified for every sequential composite that you can come up with

and every parallel composite that you can come up with.

And that's not so bad.

I'll show you,

show you,

show you one movie.

Again,

questions.

Welcome.

Is the mind a constructor?

Is the mind a constructor?

Yeah.

Well,

that depends on your,

depends on what you mean.

It depends on,

depends on the,

the,

the process theory or this,

or whatever other mathematical incarnation you pick for the constructive theory.

Remember,
it's a meta theory.
You bring your own mathematics.
So I can't answer that question until the mathematics is brought and there,
you know,
all the English has been turned into math.
And then I can say,
oh yes,
that's a short or no.
So it depends.
That's pretty good.
I like that.
Okay.
What is the,
okay.
Here,
let's say I have,
I have two tasks.
I'm not a fan of using letters.
I'm more of a fan of using colors,
but if anyone is,
if anyone is colorblind or if anything's not clear,
then just let me know.
All right.
So,
maybe there's a different,
yep.
Nice.
Then just to make sure everything matches up to this end.
Okay.
So here is how,
here is how we're going to sequentially compose tasks.
Right?
This here is a task.
task one,
task two,
both tasks,
they work on the same substrate.

So for example,

paint a shoe and then unpaint a shoe.

Sequential composition of tasks.

Well,

if you're going to sequentially compose,

it needs to be on the same substrate.

Otherwise,

sequential composition,

that's meaningless.

And you might want something like parallel composition instead.

So for example,

you can compose the tasks,

which may not be possible tasks.

So paint the shoe and then strip the shoe of paint.

I don't like it.

Do something else with it.

And here's how we will sequentially compose just conceivable tasks.

If you have a conceivable task that's operating on this orange substrate,

you need to compose something else that takes orange as that substrate again.

But what about the environment?

The environment can differ.

For example,

if I'm going to paint a shoe with paints versus if I'm going to,

I don't know,

do one of those electrolysis,

chrome dipping of a shoe to get it covered in chrome,

the environment is going to be different.

Right.

So for the first task,

I have this particular,

for the first task,

I have this particular environment.

The second task,

that has its own particular environment.

Those environments are just systems.

I say that,

well,

okay,

if I'm going to sequentially compose,

I feed in the requisite environment to do task number one.

And then I twist in and feed in the requisite environment to do task number two.

And then just to make sure that everything type checks at the end.

This entire thing here,

right.

This entire thing here.

Now,

maybe it's better if I

gather these guys here to make it clear.

Right.

That's sequential composition.

What is it saying?

Right.

You can always gather.

Systems can also be composite systems that are placed next to each other.

This entire thing says,

well,

you've got to give me the environments for the two different tasks and the substrate.

Then I'll give you back the environments again for the two different tasks and the substrate.

That's how we achieve the sequential composition of tasks.

So these two small tasks,

put them together just like this.

And then you get this one big task.

Right.

What about parallel composition?

Parallel composition looks fairly similar.

With parallel composition,

you can compose whatever.

It doesn't really,

parallel composition doesn't particularly matter.

If I get this right here.

So we've got substrate one or an orange.

We've got substrate two over here.

I'm going to do it like this.

It makes me see here.

We've got environment one.

And then environment two.

And then,

those are your two tasks.

And this is your composite task.

Is that all visible from there?

I wonder.

Hopefully that's kind of visible from this.

Yeah.

Fantastic.

So again,

what are we saying?

Well,

we've got,

if you want to parallel compose,

parallel composition of tasks is fantastic.

You don't need to match up with a substrate.

You can simultaneously on one side,

paint shoes and the other one go,

you know,

do your taxes or whatever it is.

It's another task,

whatever.

So you can have two completely different substrates.

Doesn't particularly matter.

You have two different environments for those substrates in that particular task.

And here's the parallel composition of conceivable tasks.

You know,

same principle.

If you understand sequential composition and you're happy with just intuiting the twisting wires,

then you're good.

And now to verify that these guys are,

um,

so now that to verify that the composition of possible tasks is again, a possible task.

So now let's say that these guys are possible tasks.

In other words,
that there is a particular constructor for,
uh,
green.
And there's a particular constructor.
For yellow over here.
Right.
And remember to check the condition.
Just delete those.
Right.
So how do we go about verifying this?
Well,
uh,
uh,
what we want to do is we want to apply the screen one here.
First,
we probably want some kind of formal condition on our tasks,
on our possible tasks to tell us that this goes through.
So then get us that we have this series of inequalities.
We have this series of inequalities.
I'm winging it here,
but there's going to be a proper worked out version.
The longer version of this paper.
So I'm happy to take questions while I'm just over here.
Messing around.
And then apply it again.
Right.
I think in the paper,

Well,
the principle is kind of simple.
You applied the equation that,
sorry,
the,
the containment relationship that defines possibility on,
uh,
on both of these one by one.

And you also make use of the fact that,
well,
there's the special state and that one that corresponds to
everything and delete.

You can pull those through as well,
uh,
for tasks.

Then you end up with the,

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I get to finally give a mathematically formal voice to what a constructive theorist might want to do.

So again, I have this first domain here.

What is that?

That is going to be this space of conceivable tasks.

That's going to be some symmetric monoidal category,
which is telling me this is the domain in which I'm interpreting a particular process theory.

And in this symmetric monoidal category, well, my conceivable tasks,
they can compose in sequence and in parallel.

And what do I mean by the cut?

Well, the cut that distinguishes what is possible for impossible.

Well, it turns out that the possible stuff, that is also a symmetric monoidal category.

And that's a symmetric monoidal category that embeds into the larger one.

It embeds in the same way that a subgroup embeds.

It embeds in the same way that a vector subspace or linear subspace is going to embed into a larger one.

And then everything else, well, then everything else is going to be, well, no promises.

This is not the sort of stuff, impossible stuff is the stuff that doesn't necessarily iterate instantly because they don't have constructors.

And if you've followed along so far, and there's more details in the paper,

if you want to look up the formal details,

you understand now constructor theory minus principle of locality,

which is perhaps not interesting enough to merit further comment.

But if someone's curious about it, we can talk about it.

But, you know, so what I presented so far is, you know,

a constructor theorist would also be happy to say,

yes, this is constructor theory minus principle of locality.

So you don't go around saying that this thing is just constructor theory to a constructor theorist
because they'll get mad at you.

And they'll say, no, no, no, it's missing the principle of locality.

But that's a dogma and you don't particularly need it.

Like that's, it messes a lot of things up.

Okay, so that's all I have to say for now, I think.

And yeah, happy to take it to a discussion now.

Okay.

Awesome.

Wow.

Mohamed, would you like to give a first comment or reflection or just share what angle you participated in the work from?

Yeah, I am interested in applying this formalism to the various examples that constructor theory has been applied to.

So you might have seen papers by Chiara Maletto and David on constructed theory of information and constructed theory of life.

And I'm also interested in seeing how it applies to quantum theory and if the process view can be shown to be superior to the constructed theoretic view.

Cool.

So if you have a given system of interest, like let's just say a cognitive system, like a person or an aunt or something like that, like how do you go about demarcating what tasks are in play?

Are these cognitive tasks?

Or are these the tasks that the body performs?

How do we begin to analyze a situation in a way that's amenable to being framed like this?

Okay.

So, but like this, could you refine the discourse reference?

What do you mean by this?

Yeah, if we want to understand a given informal, actual embodied system in a way that's amenable to all of the compositional, bountiful benefits, what are we actually looking to break down about that system of interest?

Sure.

So when we talked about, so tasks are kind of this ad hoc thing, right?

Remember that when we drew tasks, we drew them like directed in this way because we had to say something about direction.

And then we distinguished between their substrates and their environments.

And then we said that systems are always these things which sort of have,

sorry, the task is something that takes in the system
and the environment
and returns to the same system and environment,
which is something that you can just accept,
but also, you know, in general,
a process doesn't need to have the same inputs and output types.
So this is perfectly fine as well.

You can have a process that's shaped like that.
And maybe for some sorts of, you know,
at whatever level of abstraction you care about,
it might be more, well,
maybe there's level of abstraction
where you don't care about the actual physics of what's going on,
where everything is sort of like conservative on some level.

You might just care about a view of the system
where you have inputs and output types,
in which case the language of,
or the framework of thinking about tasks
is perhaps less fundamental.

I mean, one of the points that we're making here
is like the framework of thinking in tasks is,
a task is a constructed concept
on top of the more fundamental and primitive notion
of just being able to construct processes
or just being able to compose processes.

So then it would depend now
on what you sort of want from your notion of task.
So what we were able to show
is that look, just using the language of process theory,
we were able to code and formalize
this particular notion of what we want tasks to behave like.

But then depending on what you would consider a task to be,
maybe the same, maybe different,
you can encode it in this mathematical language
and then start reasoning about it in a diagrammatic way.
And there's a lot of things you can reason about
in process theory.

There's a laundry list of them in the paper,

like electrical circuit theory,
quantum theory, most notably,
linear algebra, first order logic.

Yeah, it's all good.

Maybe that's not quite a satisfactory direct answer,
but perhaps we can refine the question and the target.

So to add to it,

it's up to the modular or the user
to decide which processes are to be modeled.

So you can, I mean, if you want to,
if you want to think about cognitive processes,
then you are, you know,
you can create a sub-symmetric model category
to work with that.

But you can also have
more mechanical process theories,
or you can even have one
where both mechanical and cognitive processes interact.

So it's up to the model, I would say.

There's no hard and fast rule in the formalism.

It's the shoe painting factory.

But a lot to say on that.

In active inference,
we're often interested
in a general cybernetic entity or thing,
something that's engaging in sense-making
on the inbound and on action selection,
decision-making on the outbound.

And another important idea
from ecological psychology
is this notion of like extended cognition
and the environment
and the interactions of the agent
and the environment.

So I wondered, is the substrate,
you were twisting different environments in
and keeping the spotlight on the thing?

Is that how you track an entity

as things change around it?

Is by twisting in and out different contexts?

Right.

So that is just a way for...

That is just a way to define a...

To define the sequential composition

or the parallel composition of tasks

in a way that relies upon a more primitive notion

of the sequential and parallel composition of processes.

So the twisting in

and keeping a spotlight on a particular context.

So one of the things that we've seen

that we've all seen in these examples

is that you can start with a process theory

of like basic processes

that just compose in sequence and in parallel.

And then you can define a new way

to compose in sequence and in parallel

on top of that.

And that would give you

a new symmetric monoidal category

that would give you a new process theory.

So you can kind of...

Yeah, yeah.

This is process theory kind of...

There's a dog food thing here somewhere

where you're kind of eating your own dog food, right?

In the case of the twisting,

which was...

What was the twisting again?

In the case of the twisting,

we were saying that,

okay, well, look, we're going to define...

Instead of just plugging things together,

we're going to define a new kind of...

We're going to find a new kind of sequential composition

that looks like this,

where instead of just taking two boxes

and joining them together,
we do all this other business over here.
Like this is this other business over here, right?
That's allowed and good and expressive,
but that's also something
that the mathematics has no stance on.
Like if you...
So you brought in these concepts,
like this is how you're spotlighting
a particular substrate
and keep...
And switching out the context, right?
And if you look at the diagram,
it's perfectly okay
to start pointing at these things
and saying, well, of course,
you're spotlighting the substrate
and you're switching out the context,
but that's on you.
That's on you.
The mathematics just tells you,
like, this is a way to compose in sequence,
but it does so nicely enough
in a way that's...
You know, where this sort of...
We can bring our linguistic intuitions
to bear on the situation
in a much more direct way
because we have the viscosity of it available to us.
Yeah.
That's very deep.
The necessary and sufficient conveyance
is on the blackboard.
And then that enables us
to take perspectives
and direct attention
and have procedural flows
with our attention

and tell narratives
and again, have interpretations,
but the interpretation
isn't what's written on the screen.

Yes.

Yes.

Yes.

All right.

A question from the live chat.

Can you please explain
in what way does that cut,
I believe,
between the possible and impossible,
allow the modeler
to express emergent phenomena
or laws
that arise from the underlying processes
in constructor theory?

Emergent processes or laws
that arise from the underlying...

In what ways does the cut
allow the modeler
to express emergent phenomena
or laws
that arise
from the underlying processes?

Yeah.

Well,
if I could just ask
for clarification
from live chat here,
and maybe Hamza can weigh in as well,
like,
what's your...

Well, what's your favorite toy example
of the emergence?

I'm not a constructive theorist,
but, you know,

we can play with the system
and we can play with the system live
and see what we get at.
But first, we're going to...
You know,
what are you thinking
in terms of emergence?
People mean different things
sometimes when they say that, right?
How about the arising directionality
of a flock of birds?
The arising directionality
of a flock of birds?
Oh,
as in there's a dynamic system
that's evolving over time,
which is you've got these birds
that are interacting
with each other locally
and then they end up...
Yes.
I wonder.
would these...
Would these...
So,
I think the safe answer
or my intuition
would say
that
the vast majority
of things
you would consider
to be possible
are
not actually
technically
possible
in constructive theory,

right?

So,

right off the bat,

you can say,

look,

you've got this intuition.

So,

if you thought painting shoes

was a fine,

sensible thing

and you think

that birds flying around

and interacting

with each other

is a fine,

sensible thing,

that's possible.

That probably isn't

what is meant

by possible.

So,

in the case

of like a bird system,

you've got this massive

interacting system,

fine.

A constructor,

you know,

you probably could have

like a sort of

infinite shotgun

constructor

that just stops

all the birds

dead in their tracks

again and again

and you can kill

as many birds

as you want.

What would a constructor
for this sort of system be?

Well,

if you can set up
an environment
that sort of
steers the birds
a possible task,
maybe for a chaotic system,
the only possible task
is the trivial one
where you don't expect
anything to be done
and you accept
any sort of outcome.

But if you want
a particular possible task,
that's tantamount
to programming
like a flock of birds
or finding some way
to program a flock of birds
like in a deterministic
and set way
that you can keep doing it
again and again,
right?

Yeah.

So,
so what is the relationship
between constructor theory
and programming?

There's,

that's a pretty good question.

I'm going to try my best
to answer on behalf
of constructor theorists here.

But there's a sort of,
there's an information theory
or a perspective
on information theory
that also comes about
from constructor theory.
So as a,
as a math and computer scientist
myself,
I'd say that perhaps
the most fundamental
point of contact
between constructor theory
and programming
is in a notion
of information
and what in particular
you're allowed
to do with it.
So in a quantum setting,
you cannot clone information.
I mean,
your information
is carried in your qubits
and you have no cloning.
So you can't,
you can't clone information.
Whereas,
and everything's not,
everything that's not classical,
you know,
the technical terms say,
well,
everything's not quantum,
you call it classical.
In a classical setting,
you can copy information.
You can write down a one

or a zero
on a piece of paper
and you can get someone
to copy down the one
and zero
as many times as you want.
And you can stick
the ones and zeros
into filing cabinets.
And so,
and by so doing,
perform computation
with copying.
So the difference there
is,
well,
okay,
if you've got different
sort of underlying
capacities
for copying things
and not being able
to copy things
in your underlying
process theory,
what is,
I guess,
the sorts of,
there's questions
to ask about
the sorts of programs
you can write,
but that's kind of,
kind of at a,
at a much more
fundamental level.
I mean,
constructive theory,

if you,
if you gave me a,
if you gave a particular
process theory
of programs,
then we can start
talking constructive theory.
Constructive theorists,
so I think by and large
are more interested
in the physics,
which is at a much lower
and more basic level
of abstraction.
So I don't know
when you're,
so when you're asking
about like,
is this,
you know,
is this a get rich quick
scheme when it comes
to emergence?
Is this a get rich quick
scheme when it comes
to,
you know,
programming and understanding
chaotic behavior,
et cetera,
et cetera.
You know,
sorry,
but the answer is no.
Like there's,
there's nothing,
the extent,
I mean,

if you can under,
you understood the content
of constructive theory
in about like an hour
and it was all pictures,
like there's nothing
more complicated than that.
Everything else beyond
that is you hallucinating
interpretations onto
the pictures.

It's a get epistemically
rich quick scheme.
You just get a seminar.

Yeah,
yeah,
yeah,
yeah.

Get epistemically rich quick.
You know,
you've got the pictures
and they're very suggestive,
but you know,
in principle,
it's no different
from like a tarot reading.
You've got just a very
suggestive of mathematics
and you can read
whatever meanings
that you want.

And if you play
with the mathematics,
you can also get
some meanings out,
but you know,
you only get it out
as much as you put in,

in terms of like
epistemically weight
bearing interpretations
of hallucinations
in the first place,
right?

Well,
it's an interesting
date with
constructor theory
and process theory
and active inference,
none of which
are opinionated
in the last mile
about any given system,
but all of which
are providing us
higher levels
of expressivity
and increasingly
in active inference,
category theory
is being used
to represent
the kinds of
generative models
that are used.

And so there's the math
and the analytical equations
and then there's
the kinds of Bayesian graphs
that we've traditionally
looked at,
the partially observable
Markov decision processes.
And then in the last years,
there's some string diagrams

and all these other features
coming in
that are helping
like expand the scope
of what kinds
of cognitive models
and the relationships
between substrates
and cognitive processes
that we want to explore.
Indeed.

I'm familiar with
friends with Toby,
Toby Sinclair Smythe.

And yeah,
that's my only contact
to active inference,
but I can definitely see
the appeal

of having
a
non-opinionated mathematics
for things
that in the last mile,
as you say,
don't have an opinion.

But that,
you know,
I'm butchering it,
I'm tired
and I'm saying it all wrong,
but I'm saying that,
look,

I think it's a good idea
to have like a math
of composition
to deal with a complex system
that's fundamentally compositional.

Yeah.

So what do we,
what modules
do we load in
or what do we bring in
in our environment
to be able to,
to load up
and actually use?

So we said,
bring your own math.

That was right
at the beginning.

So it's like,

okay,
what do I bring?

Piano's axioms
or what do I bring?

Like my f of x
or what do you mean
by the mathematics
that has to be brought?

Okay.

So when it comes,

so,

so the,

the sort of basic,

the,

the,

the core module

that you sort of get

with process theories

is a theory of composition

that works for

anything that you can name

processes

that act on systems

and these processes

can compose
in sequence
and in parallel.
This is the,
this is the sort of basic
conceptual map
that you get
and becomes
gun to head formal
when you load it
into a process theory.

That's,
that's what you get.

That's core.

And then
everything else
is actually
expressed like
sort of in terms
of the,
in terms of
the diagram.

So for example,
this piano's axioms,
right?

Let's talk about
the,
what is it?

Nice.

I forget what the axioms are,
but I know that there are
two generators.

You've got a zero.

You've got a zero state
and then you've got
a successor box.

Right.

And maybe the first thing

that you want to do
if you want to actually,
well,
you want to load in
anything that you can express
at the end of the day
in terms of algebra
and symbols
and equations
is also doable.

With also doable,
like diagrammatically.

like for example,

I believe that

in the piano

formulation,

the addition operation

is something

that sort of

behaves like a homomorphism

with respect to

or like a module

with respect

to the successor box

or something like,

I think something like

this is right.

Yeah.

It's almost like

the axioms

are proto-procedural

in that they define

exactly what can

and can't be done.

Indeed.

Indeed.

So the sort of

benefits of thinking

in a,
or let's say
depicting
in a process
theoretic way
is that,
well,
look,
if you're more used
to talking about
processes
and how they interact
and you don't really care,
you don't want to,
you know,
spell out
from the bottom up
how they're implemented
and do a sort of
calculation
with like the specifics,
what you really
just care about
are the operational
constraints
of how they compose
and behave
when you compose
them together
and you can express
them in terms
of equations
or containment
relations
or whatever else,
then the process
theory is much better
as opposed to,

for example,
just dealing
with the sets,
you know,
especially if you've
got like a very
complex system,
then,
you know,
no hope.
Again,
to these two
key features
of cognitive systems,
like the sense making
on the inbound
and the associated
world model
and then the
action selection,
I thought about
the first blob
you drew
with the conceivable.
That's like,
well,
you drew the lines
and we have the rest
of the blackboard.
So it's like,
we know that there's
unconceivable,
but that's the scope
of the sense making
for a given setting
and then the line
between what is possible
and what isn't possible

is what is actionable
and I guess
you can develop
thought settings
where,
okay,
well,
I'm going to speculate
what could happen
if I could lift up
10,000 pounds
and then you could
do tasks
related to moving
cargo containers.
Whereas if you
were using tweezers
and you defined
what was possible
with a different threshold,
then there'd be
different kinds
of plausible tasks.
ants.
Yeah,
so,
yeah,
so for
sort of
what might interest you
is,
well,
I'm going to shill it now.
Why not?
I've got some time.
My interest to you
for this sort of reading
is that,

well,
for constructive theory,
you know,
it really just
wants to force you
on one side
of a binary
and this may be
too restrictive,
you know,
in terms of expressive capacity
when you're trying
to especially map it
onto something
that involves
anything as complex
as a distinction
between cognitive
and physical tasks
and processes,
right?
And we had questions
to this effect
earlier as well,
like what do we do
with the likely?
What do we do
with the vanishingly unlikely?
What do we do
with the plausible,
right?
How do we?
So I want to point out
that,
and this is something
that's coming
in the longer version
of the paper,

is a relationship
between
constructive
constructive theory
as a special case
of resource theories.

So,
you know,
this idea
that you have to,
with a constructive theory,
you have to either go
like infinitely many times
or,
you know,
infinite or bust.

A resource theory
is sort of going like,
yeah,
but,
you know,
we'd be happy also
if we could do it
a couple of times.

We'd be happy
if we could,
you know,
tell me how many times
you can do it.

You know,
that's pretty good.

And so,
you know,
resource theory
and principle
give you this idea
of like,
well,

you know,
here's the sort of thing
that you can do once,
you can do twice,
you can do three times,
you can do da-da-da-da-da-da.

And finally,
you know,
in the very,
very core
over here,
you know,
that's the sort of
stuff that's picked out.

That's the sort of stuff
that can be picked out
by a constructive
theoretic cut
in the limit.

Whereas a resource
theoretic point of view
is saying like,

well,
of course,
you know,
if you want to make
more distinctions
than that,
then like an easy
generalization
that's also quite simple
to think about
is let's talk about
the stuff that we can do
for free all the time.

Let's talk about the things
that we only,
you know,

we only get to do once.
And it turns out
that resource theories
are particularly,
you know,
resource theories
are actually like
a practical bit
of mathematics
when it comes to
quantum circuit
optimization
because there are
certain components
that are very expensive
and you want to be able
to work out
what sort of quantum maps
you can implement
only using this particular
say T gate,
let's say like three times
or maybe you get
like one extra one
or maybe you get
like two less
and then,
you know,
the resource theory
is going to tell you
precisely,
you know,
what your gain
and loss
in expressive capacity
is when you have
more or fewer
or more resources,

which is something
that is a very
computer science-y
down-to-earth
practical thing
as opposed to
just the distinction
between infinity
and everything else
that constructive theory
gives you.

that sneak peek
for some of those
coming,
the next installment
or the extended version
of this paper
that's going on archive.

That is super exciting
and relevant.

Certainly,
organizations
or any other kind
of system
that's being designed
and having an ability
to handle the finite,
very relevant.

Here's one other
kind of speculative question.

Of course.

We had the ice cream
melting and unmelting.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

You had the ice cream
unmelting as impossible.

Yeah.

But,
of course,
in a freezer,
in a freezer,
it does happen
and or at room temperature
with a reversed time arrow.

Yeah.

And so,
within this notion
of the Gibbs free energy,
we have catalysts
like protein enzymes
which are not able
to enable
what isn't possible,
but rather,
they can accelerate
by reducing
the activation energy
and they can accelerate
a negative ΔG
and they can accelerate
something that is possible
catalytically,
hence catalyst,
and be regenerated.

Right.

And it made me
think about
the change
in variational
free energy
as the ball
rolling downhill

cognitively
and about how
the channels
and the landscapes
of the mind
or of the cognitive system,
sometimes
the landscape
is changed
and what is possible
becomes different,
but just because
something's possible
doesn't mean
it's ever reached
and then we still
need this question
of the catalyst
to actually implement
and get it
to where it can be.

Right.

Right.

And I think
that perhaps
one of the
sort of intuitive
appeals
of constructor theory
is that,
you know,
at first glance
and with an intuitive
mapping of concepts,
this sound,
constructor theory
sounds like
the right kind

of mathematics
to talk about.
you know,
these interactive
concepts that you're
talking about,
the free energy
and catalysis
as enabling,
but the caveat
with constructor theory,
remember,
is infinite or bust,
hard binaries,
which means that
if you want to do
so,
like,
I'm sorry to say,
but if you want
to do anything
that's remotely nuanced,
more nuanced
than like
a pair of qubits
interacting,
which already,
because of principal
locality
gives constructor theorists
a lot of pain,
constructor theory
is probably
not the right
system for you.
Process theories,
though,
if you want to deal

with composition,
not bad,
and especially,
you know,
you get diagrams
for free,
and also you get
other add-ons
that are very good
at doing the sort
of things
that you want to do
that have the right
expressivity and shape,
like resource theories.

Constructive theory
one should think of
as a particular
way or methodology
that's designed
for a particular problem,
to be a metatheory
of physics
in whatever David
Deutsch considers
to be like
a sort of
conceptually clean
setting of just
being able
to make a single cut
to tell you about,
like,
to distinguish
in a black and white way
all of physics.

Interesting.

And in the Bayesian mechanics,

which is
some of the recent
developments
in active inference
and free energy
principle area,
that grounding
of physics
to,
I guess,
what we want
to call physical,
even though we know
that so many asterisks
have to be added
when we knock on the wood
and everything like that,
but if the physics
of cognitive systems
can be described
purely informationally,
then we can have
a procedural,
compositional,
cognitive physics
that would enable us
to do anything
from the mind
conceptualizing
that the shoe
could be a different color
to the multi-agent setting
and composed
across individuals
and there's just,
there's more walls
to paint on
and,

well,
there are many
possible walls
to paint on.
Yes.
Yeah.
What are you curious about
or,
I mean,
what's an exciting
direction?
In terms of,
in terms of what exactly?
Because in terms of
this whole business
with constructive theory,
this is kind of
a side project for me.
A fun one,
but just,
just a side interest.
what's your main interest
or what motivates
or what excites
the context
such that this could
be a side interest?
Well,
you know,
my main interest
is more,
I suppose it's more
aligned with
this notion of sensemaking.
my interest
is sort of
using diagrams
not only for

sensemaking
but as a formal
platform
to,
you know,
articulate
theories of what
sensemaking might be.
So I played around with,
so these diagrams,
you know,
they're good for quantum,
for example,
but the same kind of mathematics
is quite good
for other sorts
of composite,
you know,
there are other domains,
not just,
for example,
psychology
that are,
you know,
dearly in need
of some sort of
compositionally native
approach
like diagrams
like linguistics,
for example.
So,
I like to draw
language
and,
because language
is a particular
tool of sensemaking

and then to formalize
the kind of sensemaking
that we make.
For example,
when we,
you know,
we use,
we use metaphor
to like,
to get the point across
relies on a complex metaphor
where information
is stuck into
words as a box
and then the box
is moved across a conduit
and this sort of sensemaking
is beyond the merely,
let's say,
truth theoretic,
yet nevertheless
is systematic
in a way
such that
the compositional aspects
of the language
precisely map onto
the compositional aspects
of the domain
of sensemaking.
And I suspect
that's where
we might have more
to talk about
because,
you know,
sensemaking is,
I consider sensemaking,

it's,
you know,
capital S,
very,
it's a,
it's a very big thing
and perhaps like
the important activity
to understand.
And I'm interested
in the systematicity
of it
and in particular
trying to depict
that systematicity.
There's something,
there's something
that's,
you know,
direct
about the,
about string diagrams
in the same way
that perfume
or music
is sort of directed,
it's not mediated
by the,
you know,
by the,
by the rows of Greek
and the symbols,
right?
So,
I don't know,
maybe,
maybe that's,
yeah.

Wow.

There's a lot there.

Can we say

that the,

the screen

is the substrate

that we're writing

language on

if we're going

by hand

with pen

and then we have

an infinite pen

and this is,

this is the inexhaustible pen

and then

that certainly

does seem

to be one thing

at which the,

yeah,

yeah.

Yeah.

I mean,

you mean to refer to

in the context

of the mind,

we have a,

I mean,

as long as we're not

aphantasic,

we have a sort of

theater of mind

that sort of

infinitely refreshable

paper

or infinitely refreshable

stage

for us to arrange
things on
and,
and sort of
replenish
the representation.
That's,
that does sound
kind of like
a constructor
if we're,
if we're happy
also to sort
sort of rule out
Heraclitus
or Heraclitus.
You mentioned
Heraclitus earlier,
you know,
man never steps
to the same river
twice,
you know,
are we,
insofar as we are
the same entity
that is perhaps
observing the theater
of mind
as we're
intranetally refreshing
it,
then fine,
yeah,
it's a constructor,
but is it the same
if,
you know,

10 years on
you're,
you're a functionally
different person,
is that still,
uh,
is it,
is it still
the same task?
If it's,
if it's,
is it still the same
task refreshing
or looking at a memory
10 years on?

I don't know.

The eye
is able to,
to nest
subtasks,
however,
it may not have
the awareness
to know what it is
embedded within,
but at least
within this
internal theater,
it can make
a sub
representation
that it
believes
could be wrong,
but that it
believes
would have
continuity

also beyond
itself,
but also
not saying
what is beyond
itself.

Beyond
itself

in,
in,
in what
sense?

Um,
like
there's the
environments
that,

um,
one is
confronted
with

and then
one could
at least
suspect
that there
are other
environments
that one

is not
being
confronted
with,

but also
know
that they
can't say
anything more

than that
about
environments
that they
aren't
being
confronted
with.

yes,

I,

I,

I,

I,

I'm not

sure what

to say to

that,

but I'll

have to

think about

that.

Mohamed,

any last

comments

before you have

to go?

Um,

well,

I'm,

I'm pretty

good.

Um,

just,

I'd like

everyone to

know that

the,

the,

the longer
version of
this very
paper is,
uh,
going to
come out
soon-ish
on Archive
where,
uh,
we work out
some more
examples,
we relate
to constructor
theory to
resource
theories,
and everything
is going
to be
very readable.
So if this
condensed,
uh,
journal version
is a bit
technical and
difficult to
follow,
watch the
space and
there's going
to be a,
uh,
longer and
more pedestrian

version that
hopefully everyone
can follow.

Yeah.

Great.

Thank you both
for joining.

Hope to,
um,
see you again.

It's a very
exciting and
evocative line
of work.

Thank you.

Thanks,

Daniel.

Hope to see
you again
soon.

Farewell.

Bye.

Bye.

Bye.

Thank you.